Megalencephalic Leukoencephalopathy With Subcortical Cysts 2B (MLC2B)

Disease: Megalencephalic Leukoencephalopathy With Subcortical Cysts, type 2B (remitting, with or without intellectual disability) **MONDO:** MONDO:0013491 · **OMIM:** 613926 · **Causal gene: HEPACAM** (encoding **GlialCAM / hepaCAM**, HGNC:17027) · **Category:** Mendelian (autosomal **dominant**)

Summary

Megalencephalic leukoencephalopathy with subcortical cysts type 2B (**MLC2B**) is a rare, autosomal **dominant**, **remitting** leukoencephalopathy caused by a single heterozygous (usually missense) variant in **HEPACAM**, the gene encoding the astrocyte-enriched immunoglobulin-superfamily cell-adhesion molecule **GlialCAM** (also called hepaCAM). It sits within the "megalencephalic leukoencephalopathy with subcortical cysts" (MLC) disease family, which also comprises classic recessive **MLC1** (MLC1 gene) and recessive **MLC2A** (biallelic HEPACAM). HEPACAM variants account for roughly **25% of all MLC cases**; the dosage/zygosity of the HEPACAM lesion determines whether the course is the deteriorating classic phenotype (two mutations → MLC2A) or the improving/remitting phenotype (one dominant mutation → MLC2B) (PMID: 21419380).

Mechanistically, GlialCAM is the obligatory trafficking partner of the astrocytic membrane protein **MLC1** and an **auxiliary (β) subunit of the ClC-2 chloride channel**, targeting the MLC1–ClC-2 complex to cell–cell junctions at perivascular astrocyte endfeet of the **gliovascular unit**. Dominant MLC2B mutations cluster in the first immunoglobulin (Ig) domain and perturb GlialCAM–GlialCAM self-interactions (cis/trans), acting in a **dominant-negative** manner to mislocalize the complex, disrupt astrocytic **volume-regulated anion channel (VRAC)** activity and ion/water homeostasis, and thereby produce chronic white-matter edema with intramyelinic vacuolation (PMID: 22405205; PMID: 31960914). Because a single dominant allele causes only a partial, developmentally timed disturbance, the edema and MRI abnormalities characteristically **improve or normalize over time**, distinguishing MLC2B from the progressive recessive forms.

Clinically, MLC2B presents in infancy with **macrocephaly** and MRI white-matter edema ± subcortical cysts, but—unlike classic MLC—motor and cognitive function are typically preserved or only mildly affected, and imaging improves. The phenotypic spectrum ranges from **benign familial macrocephaly** (full normalization) to **macrocephaly with intellectual disability ± autism** (PMID: 21419380). There is **no disease-modifying therapy**; management is supportive. This report synthesizes seven confirmed findings and 23 reviewed papers into a comprehensive, section-by-section knowledge-base entry.

Key Findings

Finding 1 — MLC2B is caused by a single dominant HEPACAM (GlialCAM) mutation and follows a remitting course

López-Hernández et al. (2011) identified HEPACAM/GlialCAM as a direct MLC1-binding partner and screened 40 MLC patients who lacked MLC1 mutations. Of these, **10 patients with the classical, deteriorating phenotype carried two HEPACAM mutations** (recessive **MLC2A**), whereas **18 patients with an improving phenotype carried a single mutation** (dominant **MLC2B**). Most parents carrying a single mutation themselves had macrocephaly, confirming **dominant inheritance with a milder, self-improving expression**. Overall, approximately **25% of MLC cases carry HEPACAM variants**.

> *"Ten patients with the classical, deteriorating phenotype had two mutations, and 18 patients with the improving phenotype had one mutation. Most parents with a single mutation had macrocephaly, indicating dominant inheritance."* — PMID: 21419380

The OMIM designation, autosomal dominant inheritance, and remitting character are reaffirmed in the recent literature:

> *"megalencephalic leukoencephalopathy with subcortical cysts 2B (MLC2B, MIM# 613,926, autosomal dominant)"* — PMID: 38280046

This establishes the central genotype–phenotype rule of the HEPACAM-linked MLC family: **allele dosage dictates course** — one dominant allele → remitting MLC2B; two alleles → progressive MLC2A.

Finding 2 — GlialCAM is an auxiliary subunit of the ClC-2 chloride channel and of MLC1; mutations mislocalize the complex

Jeworutzki et al. (2012) demonstrated that GlialCAM physically binds the **ClC-2 chloride channel**, co-localizes with it in **Bergmann glia and perivascular astrocyte endfeet**, and **targets ClC-2 to cell–cell junctions**, where it increases ClC-2-mediated currents and alters the channel's gating. Crucially, disease-causing GLIALCAM mutations **abolish this junctional targeting**, which is the core molecular lesion shared across MLC2A and MLC2B.

> *"GlialCAM targets ClC-2 to cell junctions, increases ClC-2 mediated currents, and changes its functional properties. Disease-causing GLIALCAM mutations abolish the targeting of the channel to cell junctions."* — PMID: 22405205
>
> *"the disruption of the Cl(-) channel ClC-2 causes fluid accumulation leading to myelin vacuolation"* — PMID: 22405205

Arnedo et al. (2014) independently confirmed that GlialCAM acts as the auxiliary (β) subunit of **both MLC1 and ClC-2**. This defines GlialCAM as a molecular hub whose job is to assemble and correctly localize an astrocytic ion-homeostasis module; mislocalization of that module produces the fluid accumulation and myelin vacuolation that are the histopathological hallmarks of MLC.

Finding 3 — Structural basis distinguishing dominant (MLC2B) from recessive (MLC2A) GLIALCAM mutations

Elorza-Vidal et al. (2020) built a structural model of GlialCAM homo-interactions and proposed a mechanistic explanation for why some HEPACAM mutations are dominant. **Dominant (MLC2B) mutations affect GlialCAM–GlialCAM interacting surfaces in the first immunoglobulin (Ig) domain**, which mediate self-association **in cis** (between GlialCAM molecules in the same cell) and **in trans** (between GlialCAM on neighbouring cells). Because GlialCAM self-associates, a mutant monomer can poison wild-type partners — the molecular definition of a **dominant-negative** effect — explaining dominant inheritance even though most pathogenic mutations (dominant or recessive) ultimately disrupt junctional targeting.

> *"we suggest that dominant mutations affect different GlialCAM-GlialCAM interacting surfaces in the first Ig domain, which can occur between GlialCAM molecules present in the same cell (cis) or present in neighbouring cells (trans)"* — PMID: 31960914

This is directly supported by functional case data: a novel MLC2B mutation **p.Q56P** mislocalized GlialCAM and its partners yet still **activated ClC-2 like wild-type**, whereas a recessive MLC2A allele (p.R73W) showed no trafficking defect and no functional change — illustrating that the dominant/recessive distinction is encoded in *which* molecular surface is disturbed rather than simply the presence of a mutation (PMID: 24202401).

Finding 4 — Clinical phenotype: infantile macrocephaly with transient white-matter edema that improves over time

The defining clinical feature of MLC2B is **clinical and MRI improvement/normalization over time**, contrasting with the deterioration of classic/recessive MLC. The multi-institutional cross-sectional study by Hamilton et al. (2018; 204 classic MLC patients) established that clinical and MRI features alone do not distinguish subtypes at a single time point, but **course over time** does (PMID: 29661901). López-Hernández (2011) delineated the MLC2B spectrum:

> *"In some families with dominant HEPACAM mutations, the clinical picture and magnetic resonance imaging normalized, indicating that HEPACAM mutations can cause benign familial macrocephaly. In other families with dominant HEPACAM mutations, patients had macrocephaly and mental retardation with or without autism."* — PMID: 21419380

The remitting/improving white-matter course has been documented serially on MRI, including in HEPACAM-region deletion (Jacobsen syndrome) cases that mimic MLC2B remission:

> *"the WMA in the present patient improved over time. As with some of MLC2B patients, the WMA in the present patient improved over time."* — PMID: 32507665

Resolution of MRI abnormalities over childhood has also been captured in teaching neuroimaging reports of MLC (PMID: 24821940).

Finding 5 — MLC converges on astrocytic VRAC dysfunction, with an emerging GPRC5B signaling node

Recent work places HEPACAM/GlialCAM within an astrocytic **volume-regulated anion channel (VRAC)** module. Kerst et al. (2026) state that genetic defects in astrocytic VRAC-modulating proteins (**MLC1, GlialCAM, Aquaporin-4, GPRC5B**) cause MLC, and disrupted VRAC activity in MLC-patient lymphoblasts and primary astrocytes supports this pathogenic link. Importantly, however, astrocyte-specific **Lrrc8a** (the pore-forming VRAC subunit) knockout **alone did not reproduce chronic edema**, refining — rather than contradicting — the model: VRAC loss is part of, but not sufficient for, the full MLC phenotype.

> *"Genetic defects in astrocytic VRAC modulating proteins (MLC1, GlialCAM, Aquaporin-4, GPRC5B) cause the leukodystrophy Megalencephalic leukoencephalopathy with subcortical cysts (MLC), characterized by chronic white matter edema and myelin vacuolization."* — PMID: 41654199
>
> *"astrocyte specific Lrrc8a knockout mice had normal brain water content, no myelin vacuolization, and preserved expression of MLC-related proteins"* — PMID: 41654199

A parallel biochemical line of evidence identifies the orphan G-protein-coupled receptor **GPRC5B** as a signaling node regulated by GlialCAM/MLC1, with dominant GPRC5B mutations found in MLC patients — introducing a **gain-of-function signaling** dimension downstream of the adhesion complex:

> *"GlialCAM and MLC1 may regulate cell signaling via GPRC5B"* — PMID: 41314544

Finding 6 — MLC is a developmental disorder of the gliovascular unit; dominant hepaCAM mutants mislocalize and rewire the astrocyte interactome

Gilbert et al. (2021) established that during postnatal development, MLC1 forms a membrane complex with GlialCAM at the **junctions between perivascular astrocytic processes**, which together with blood vessels constitute the **gliovascular unit** — framing MLC as a **developmental disorder of this unit**.

> *"MLC1 forms a membrane complex with GlialCAM (another astrocytic transmembrane protein) at the junctions between perivascular astrocytic processes. Perivascular astrocytic processes along with blood vessels form the gliovascular unit."* — PMID: 34723793

Lewis et al. (2025/2026), studying three **dominant** pathogenic hepaCAM variants in the developing mouse cortex *in vivo*, found that mutant hepaCAM is **dramatically redistributed** throughout cortical astrocytes and **significantly rewires the protein interactome** — with **decreased association with Connexin 43 and ClC-2** (coupling/ion-handling partners) and a **novel interaction with the epilepsy-associated potassium channel KCNQ2**:

> *"We also observed significant changes in protein interactome between wild type and mutant hepaCAM, including decreased association with previously described hepaCAM-interacting proteins Connexin 43 and CLC-2. Moreover, we identified the epilepsy-associate potassium channel KCNQ2 as a novel hepaCAM interaction partner"* — PMID: 42649352

This mechanistically links dominant hepaCAM mutations to both **gap-junctional coupling defects** (Connexin 43) and **seizure susceptibility** (KCNQ2), and emphasizes a cell-biological, developmentally timed mechanism consistent with the remitting course.

Finding 7 — GlialCAM operates in an adhesion/signaling axis with MLC1 and aquaporin-4 controlling astrocyte state and water handling

De et al. (2023), using human glioblastoma specimens and single-cell transcriptomics, independently validated functional links among **GlialCAM, Mlc1, and aquaporin-4 (AQP4)**. High GlialCAM promotes cell–cell adhesion, while its loss activates pro-invasive extracellular-matrix adhesion/signaling. Although this was a cancer-biology study, it independently confirms GlialCAM's normal astrocyte adhesion role and its partnership with the **water channel AQP4** — the same axis disrupted in MLC2B.

> *"Profiling GlialCAM-regulated genes combined with cross-referencing to single-cell transcriptomic datasets validates functional links among GlialCAM, Mlc1, and aquaporin-4 in the invasive cell state."* — PMID: 37722850

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous dominant missense variant in HEPACAM** (clustering in the first Ig domain) **leads to** production of a mutant GlialCAM monomer alongside wild-type protein. *(Demonstrated — López-Hernández 2011; Elorza-Vidal 2020.)*
2. The mutant monomer **disrupts GlialCAM–GlialCAM self-association** (cis within a cell and trans between neighbouring astrocytes), **resulting in** a **dominant-negative** impairment of GlialCAM assembly. *(Inferred from structural modelling — Elorza-Vidal 2020.)*
3. Impaired GlialCAM self-assembly **results in** failed **targeting of the GlialCAM–MLC1–ClC-2 complex to cell–cell junctions** at perivascular astrocyte endfeet. *(Demonstrated — Jeworutzki 2012; case p.Q56P, Arnedo 2014.)*
4. Mislocalization **leads to** rewiring of the astrocyte interactome: **decreased association with Connexin 43 (gap-junction coupling) and ClC-2**, and **aberrant interaction with KCNQ2**. *(Demonstrated in mouse cortex in vivo — Lewis 2025/2026.)*
5. These coupling/channel defects **result in** disturbed astrocytic **ion (Cl^-, K^+) and water handling**, including impaired **volume-regulated anion channel (VRAC)** activity and dysregulated **AQP4**-mediated water flux. *(Demonstrated/supported — Kerst 2026; De 2023; Mlc1-null mice.)*
6. Impaired astrocyte **regulatory volume decrease leads to astrocytic swelling** at fluid–brain barriers (the earliest change in model mice). *(Demonstrated — Mlc1-null mice,* *.)*

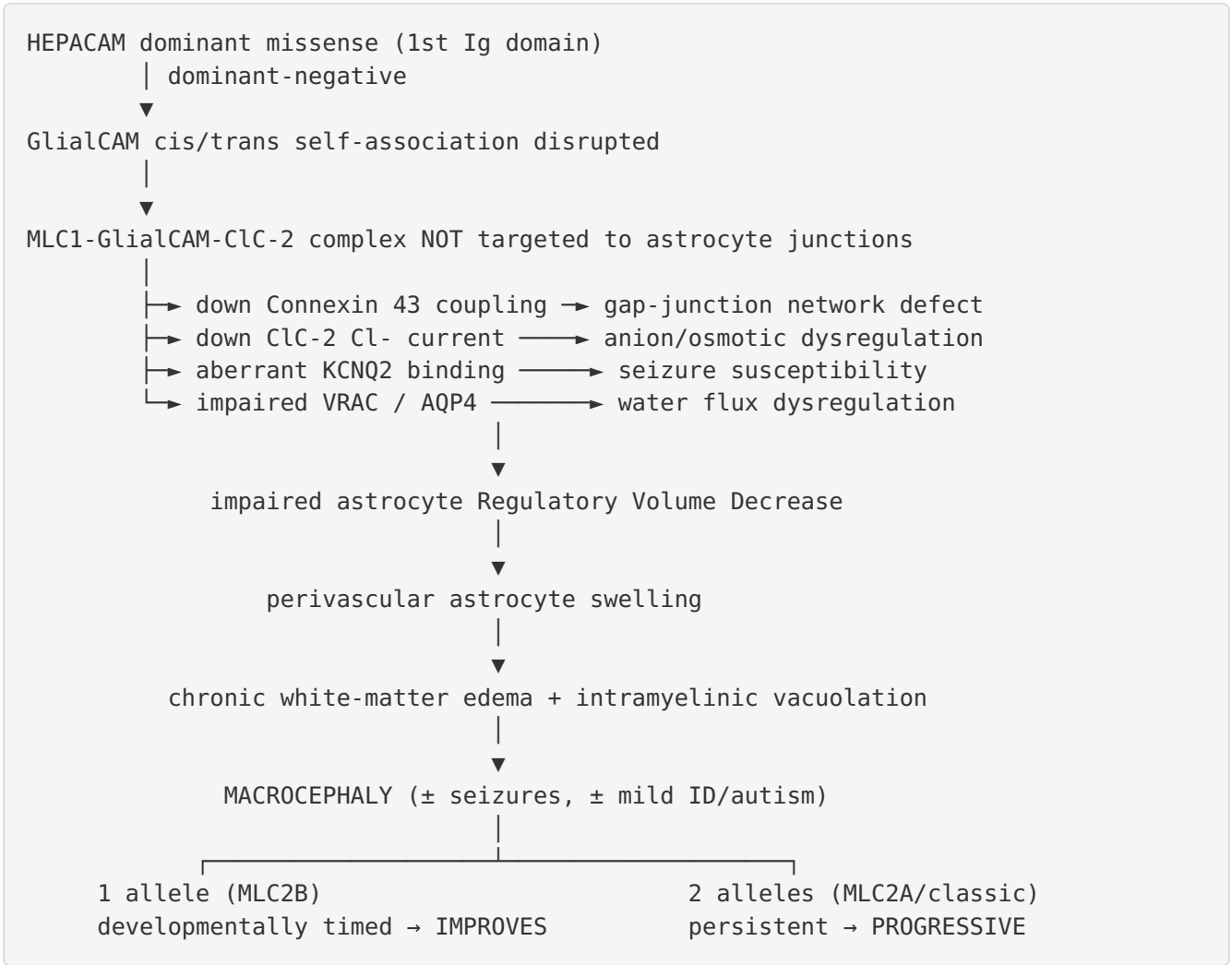
   P *25382142*
7. Astrocyte swelling and fluid dysregulation **result in chronic white-matter edema with intramyelinic (myelin) vacuolation** and, in classic forms, **subcortical cysts**. *(Demonstrated histopathology.)*
8. Edema **manifests clinically** as **infantile macrocephaly**, ± seizures, ± mild motor/cognitive involvement. *(Demonstrated clinically.)*
9. **Branch (MLC2B-specific):** Because only **one** dominant allele is affected, the homeostatic disturbance is **partial and developmentally timed** — as MLC1/GlialCAM expression peaks in infancy and then declines, the defect **leads to spontaneous improvement/ normalization** of edema and MRI, producing the **remitting** MLC2B phenotype (benign familial macrocephaly ↔ macrocephaly with ID/autism). *(Inferred from genotype–phenotype*

and developmental expression data — López-Hernández 2011; *.)*

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10. **Branch (MLC2A / classic MLC): Two** HEPACAM alleles (or biallelic MLC1 loss) **lead to** a more complete, **persistent** homeostatic failure → **progressive** deterioration rather than remission. *(Demonstrated.)*
11. **Emerging parallel node:** GlialCAM/MLC1 regulate the orphan GPCR **GPRC5B**; dominant GPRC5B mutations can themselves cause MLC, suggesting a **gain-of-function signaling** contribution acting alongside the adhesion/ion-homeostasis defect. *(Emerging — Kerst 2026.)*

Schematic



Upstream vs downstream

- **Upstream (initiating):** HEPACAM dominant mutation → GlialCAM self-assembly/trafficking failure.
- **Midstream (effector):** mislocalization of MLC1–ClC-2; loss of Connexin 43 coupling; VRAC/AQP4 dysregulation.
- **Downstream (tissue/clinical):** astrocyte swelling → myelin vacuolation/edema → macrocephaly, seizures, variable cognition.

Cell types and biological processes (ontology suggestions)

- **Cell types (CL):** astrocyte (**CL:0000127**); perivascular astrocyte / astrocyte endfoot; Bergmann glia (**CL:0000644**); oligodendrocyte (secondarily, via myelin vacuolation, **CL:0000128**).
- **Biological processes (GO):** cell volume homeostasis (**GO:0006884**); regulation of regulatory volume decrease; chloride transmembrane transport (**GO:1902476**); water transport (**GO:0006833**); cell–cell adhesion (**GO:0098609**); gliogenesis / astrocyte development (**GO:0014002**).
- **Cellular components (GO CC):** plasma membrane (**GO:0005886**); cell junction (**GO:0030054**); anchoring junction; astrocyte endfoot plasma membrane; myelin sheath (**GO:0043209**).

Section-by-Section Knowledge Base Entry

1. Disease Information

MLC2B is a rare genetic vacuolating leukoencephalopathy — the **dominant, remitting** HEPACAM form of the MLC family. **Identifiers:** OMIM **613926**; MONDO **MONDO:0013491**; gene **HEPACAM** (HGNC:17027, GlialCAM). Orphanet classifies MLC under ORPHA:2478 (MLC) with gene-defined subtypes. **Synonyms:** "megalencephalic leukoencephalopathy with subcortical cysts 2B, remitting, with or without mental retardation"; "MLC2B"; dominant-HEPACAM MLC. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet) and **published patient cohorts/case series** (e.g., López-Hernández 2011; Hamilton 2018), not EHR.

2. Etiology

Primary cause: a single heterozygous (dominant) pathogenic variant in **HEPACAM**, predominantly **missense** variants affecting the first Ig domain, acting **dominant-negatively** (PMID: 21419380; PMID: 31960914). **Genetic risk factors:** the HEPACAM variant itself is causal; **MLC1** and **GPRC5B** are related genes within the same disease family / pathway. **Modifier/ allele-dosage effect:** zygosity is the principal modifier of course (1 allele → remitting MLC2B; 2 alleles → progressive MLC2A). **Environmental risk/protective factors and gene–environment interactions:** none established — MLC2B is a monogenic Mendelian disorder with no known environmental contribution. **No infectious etiology.**

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Course	Frequency
Macrocephaly	Physical sign	HP:0000256	Infantile	Persistent (even after MRI improves)	Nearly all
White-matter edema / abnormal MRI signal	Imaging/ lab	HP:0002500 / HP:0040146	Infantile	**Improving/ remitting** in MLC2B	Characteristic
Subcortical cysts	Imaging	HP:0002416 / HP:0006891	Infantile	May regress	Variable (less than classic)
Seizures / epilepsy	Clinical sign	HP:0001250	Childhood	Variable	Subset
Intellectual disability	Cognitive	HP:0001249	Childhood	Mild/variable	Subset ("with or without")
Autism / autistic features	Behavioral	HP:0000717	Childhood	Variable	Subset
Motor delay / mild motor deterioration	Clinical	HP:0001270	Childhood	Usually mild in MLC2B	Subset

Quality-of-life impact: at the **benign end**, essentially normal functioning (benign familial macrocephaly); at the **more affected end**, impact is driven by intellectual disability, autism, and/or seizures rather than by progressive motor decline, which distinguishes MLC2B from classic MLC.

4. Genetic / Molecular Information

Causal gene: HEPACAM / GlialCAM (chromosome 11q24.2; OMIM gene 611642). **Variant classes:** predominantly **missense** (dominant MLC2B); nonsense/frameshift and compound-heterozygous combinations, and an intragenic duplication, are seen in recessive MLC2A (PMID: 24202401; PMID: 38280046). **Representative dominant allele:** p.Q56P (MLC2B; mislocalizes GlialCAM yet preserves ClC-2 activation) (PMID: 24202401). **Classification:** per ACMG/AMP, recurrent MLC-associated missense variants are typically pathogenic/likely pathogenic (ClinVar). **Allele frequency:** pathogenic variants are rare/absent in gnomAD. **Origin:** germline; dominant variants are frequently **de novo** or inherited from a mildly affected/macrocephalic parent. **Functional consequence: dominant-negative** disruption of GlialCAM self-assembly and complex trafficking (MLC2B); **loss of function** in biallelic MLC2A. **Modifier genes:** allele dosage; pathway partners MLC1, GPRC5B, AQP4. **Chromosomal:** large deletions encompassing HEPACAM (11q terminal deletion / Jacobsen syndrome) can produce an MLC2B-like remitting white-matter picture (PMID: 32507665).

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious factors are known to cause or modify MLC2B. It is a purely monogenic disorder.

6. Mechanism / Pathophysiology

Presented in full above (see **Mechanistic Model**). Summary: dominant HEPACAM mutation → GlialCAM self-assembly/trafficking failure → mislocalized MLC1–ClC-2 complex, reduced Connexin 43 coupling, aberrant KCNQ2 binding, impaired VRAC/AQP4 water handling → astrocyte swelling → white-matter edema/vacuolation → macrocephaly ± seizures ± mild cognitive involvement; partial single-allele defect and developmental timing yield the remitting course. **Molecular pathway:** astrocytic ion/water homeostasis (ClC-2 chloride transport, VRAC/LRRC8A, AQP4 water channel), GlialCAM/MLC1 adhesion, GPRC5B signaling. **Cellular processes:** regulatory volume decrease, gap-junctional coupling, cell adhesion. **Protein dysfunction:** dominant-negative mislocalization (not aggregation). **Immune**

involvement: none causal, though anti-GlialCAM autoantibodies appear in multiple sclerosis (separate context, PMID: 35628339). **Tissue damage mechanism:** osmotic/fluid accumulation → intramyelinic vacuolation.

7. Anatomical Structures Affected

- **Organ/system:** central nervous system (**UBERON:0001017**); cerebral **white matter** (**UBERON:0002316**) is the primary target; brain (**UBERON:0000955**).
- **Preferential sites:** cerebral hemispheric white matter with **anterior-temporal and frontoparietal subcortical cysts**; perivascular astrocyte endfeet (gliovascular unit).
- **Tissue/cell level:** nervous tissue; **astrocytes** (CL:0000127), especially perivascular astrocytes and **Bergmann glia** (CL:0000644); secondary **myelin/oligodendrocyte** involvement.
- **Subcellular (GO CC):** plasma membrane, **cell junctions** (GO:0030054), astrocyte endfoot membrane, myelin sheath (GO:0043209).
- **Lateralization: bilateral and symmetric** white-matter involvement.

8. Temporal Development

- **Onset:** infantile (macrocephaly often noted in first year); **insidious/chronic** onset.
- **Course (MLC2B hallmark): remitting/improving** — clinical and MRI abnormalities frequently stabilize, improve, or normalize over childhood, in contrast to the slowly progressive deterioration of classic/recessive MLC (PMID: 21419380; PMID: 29661901; PMID: 32507665).
- **Remission pattern:** spontaneous (developmentally timed), not treatment-induced.
- **Critical period:** infancy/early childhood, coinciding with peak astrocytic MLC1/GlialCAM expression and myelination.

9. Inheritance and Population

- **Inheritance:** autosomal **dominant** (MLC2B); contrast recessive MLC1/MLC2A.
- **Penetrance/expressivity: incomplete penetrance and highly variable expressivity** — ranges from asymptomatic/benign familial macrocephaly to macrocephaly with intellectual disability ± autism within the same family (PMID: 21419380).
- **De novo vs inherited:** both; mildly affected macrocephalic parents may transmit.

- **Epidemiology:** MLC overall is very rare (orphan disease); HEPACAM accounts for ~25% of MLC, of which MLC2B is the dominant subset. Precise prevalence/incidence of MLC2B specifically is not established; higher MLC frequency is reported in consanguineous and certain founder populations — these largely reflect **recessive** forms.
- **Sex ratio:** no strong sex bias reported. **Consanguinity:** relevant to recessive forms, not dominant MLC2B.

10. Diagnostics

- **Imaging (primary):** brain **MRI** showing diffuse, bilaterally symmetric white-matter signal abnormality/edema with anterior-temporal (± frontoparietal) subcortical cysts; **serial MRI showing improvement** supports MLC2B (PMID: 24821940; PMID: 32507665).
- **Clinical clue:** infantile macrocephaly (serial head-circumference).
- **Genetic testing (confirmatory): single-gene HEPACAM sequencing** or a **leukodystrophy/MLC gene panel** (HEPACAM + MLC1); **WES/WGS** when panels are negative; **CMA** can detect 11q deletions mimicking MLC2B. Zygosity determination (one vs two HEPACAM alleles) distinguishes MLC2B from MLC2A.
- **Biopsy/pathology (historical/research):** myelin and perivascular astrocyte **vacuolation** — rarely needed clinically.
- **No specific blood/urine biomarker or enzyme assay** exists; diagnosis is imaging + genetic.
- **Differential diagnosis:** classic MLC (MLC1, recessive), Canavan disease, vanishing white matter disease, Alexander disease, Jacobsen syndrome (11q deletion), other megalencephalic/vacuolating leukodystrophies — distinguished by causal gene, cyst distribution, and especially **course** (remitting vs progressive).

11. Outcome / Prognosis

- **Prognosis is comparatively favourable** for MLC2B: motor and cognitive function are typically preserved or only mildly affected, and imaging/clinical features improve (PMID: 21419380).
- **Life expectancy:** not reported to be reduced in the benign end of the spectrum; no disease-specific mortality data.
- **Morbidity:** driven by residual macrocephaly, variable intellectual disability/autism, and seizures (subset), rather than progressive disability.
- **Prognostic factors:** allele dosage (dominant single allele → better), presence/absence of seizures and cognitive involvement.

12. Treatment

- **No disease-modifying therapy exists;** management is **supportive/symptomatic** (PMID: 38487253; PMID: 33551753).
- **Supportive care (NCIT suggestions):** antiepileptic/anticonvulsant drugs for seizures; developmental/educational support, physical/occupational/speech therapy for those with motor or cognitive involvement; behavioral support for autism features.
- **No approved gene, cell, RNA, or targeted therapy** for MLC2B. Experimental directions target the shared MLC pathway (e.g., VRAC/ion-homeostasis modulation); none is MLC2B-specific. (Note: guanabenz in the PMID: 42586097 trial is for **vanishing white matter**, a different leukodystrophy — not MLC2B.)

13. Prevention

- **No primary prevention** (monogenic disease). **Genetic counseling** is the principal intervention: autosomal dominant with ~50% transmission risk per child from an affected parent, but with **incomplete penetrance/variable expressivity** complicating prediction.
- **Prenatal / preimplantation genetic testing** is possible once the familial HEPACAM variant is known. **Cascade testing** of relatives (noting many carriers have only macrocephaly) is appropriate.
- No immunization, behavioral, or public-health prevention applies.

14. Other Species / Natural Disease

- **Orthologs:** mouse *Hepacam* / *Mlc1* (NCBI Gene; model species *Mus musculus*, NCBI:txid10090). GlialCAM/MLC1 biology is evolutionarily conserved across mammals.
- **Natural disease:** no well-characterized naturally occurring HEPACAM-MLC2B equivalent in companion animals is established in the reviewed literature (OMIA not confirmed here).
- **No zoonotic or cross-species transmission** (genetic disease).

15. Model Organisms

- **Mouse — Mlc1-null:** recapitulates early human disease (early-onset megalencephaly, increased brain water content, abnormal perivascular astrocytes from ~3 weeks, widespread white-matter vacuolization/intramyelinic edema from ~3 months); astrocytes show **slowed regulatory volume decrease and reduced volume-regulated anion currents**, rescued by MLC1 re-expression, and **reduced GlialCAM and ClC-2 expression** — confirming the astrocyte volume-regulation defect as the primary lesion (PMID: 25382142).

- **Mouse — GlialCAM / hepaCAM models:** developmental in-vivo studies of dominant hepaCAM variants show redistribution and interactome rewiring in cortical astrocytes (PMID: 42649352, PMID: 40894676).
- **Mouse — Lrrc8a (VRAC) conditional KO:** astrocyte-specific deletion causes neurological dysfunction but **not** chronic white-matter edema — a useful negative/refining model (PMID: 41654199).
- **In vitro:** patient lymphoblasts and primary astrocytes (VRAC assays); heterologous expression systems (Xenopus oocytes / HEK/HeLa) for ClC-2 trafficking and current studies (PMID: 22405205).
- **Model limitation:** most genetic models address MLC1 loss (recessive) rather than the **dominant-negative** MLC2B mechanism, and models do not fully capture the **remitting** human course.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
21419380	Mutant GlialCAM causes MLC, benign familial macrocephaly, macrocephaly+retardation/autism	Human clinical/ genetic	F1, F4 — dominant 1-allele → remitting MLC2B; phenotype spectrum
22405205	GlialCAM is a ClC-2 auxiliary subunit	In vitro / molecular	F2 — ClC-2 β-subunit, junctional targeting, mutations abolish it
31960914	Structural basis of dominant vs recessive GLIALCAM mutations	Computational/ structural	F3 — dominant = 1st Ig domain self-interaction surfaces
24202401	Expanding spectrum: two GLIALCAM patients (MLC2B p.Q56P; MLC2A)	Human + in vitro	F3 — dominant vs recessive allele functional contrast
29661901	MLC disease variant characterization (204 patients)	Human cross-sectional	F4 — course distinguishes subtypes
32507665	Serial MRI in Jacobsen syndrome (HEPACAM deletion)	Human imaging	F4 — remitting white-matter course
24821940	Resolution of MRI abnormalities in MLC	Human imaging	F4 — MRI improvement
38280046	Intragenic HEPACAM duplication → MLC2A	Human genetic	F1 — OMIM 613926, AD; HEPACAM allelic series
41654199	Astrocyte-specific Lrrc8a KO	Mouse model	F5 — VRAC module; LRRC8A loss insufficient
41314544	GPRC5B regulated by MLC1/GlialCAM	In vitro/biochem	F5 — GPRC5B signaling node
34723793	MLC as developmental disorder of the gliovascular unit	Review/model	F6 — gliovascular-unit localization
42649352	Dominant hepaCAM variants alter localization & interactome	Mouse in vivo	

PMID	Title (abbrev.)	Evidence type	Supports
			F6 — interactome rewiring (Cx43, ClC-2, KCNQ2)
40894676	Dominant MLC hepaCAM mutations alter localization/interactome	Mouse in vivo	F6 — ~25% MLC from hepaCAM; dominant mechanism
37722850	GlialCAM in glioblastoma cell state	Human/scRNA-seq	F7 — GlialCAM–MLC1–AQP4 axis
25382142	Mlc1-null mice: developmental angle	Mouse model	Mechanism — astrocyte RVD/VRAC defect, ↓GlialCAM/ClC-2
38487253	MLC variant update & review	Review	Treatment — "No treatment is available"
33551753	MLC pathophysiology & therapy perspectives	Review	Mechanism, treatment
25883547	MLC1 links leukodystrophies & channelopathies	Review	Mechanism — ion/water homeostasis
35628339	GPR37 & MLC; anti-GlialCAM in MS	Review	Immune context (separate)

Consistency: All seven confirmed findings are mutually reinforcing. The dominant-negative structural model (F3) explains the dominant inheritance established clinically (F1); the ClC-2/MLC1 trafficking role (F2) and gliovascular-unit localization/interactome data (F6) explain the ion-water homeostasis defect; the VRAC and AQP4 convergence (F5, F7) and Mlc1-null mouse data unify MLC-family genes onto one astrocytic volume-regulation module. No reviewed paper contradicts the core model; the Lrrc8a-KO negative result (F5) refines rather than refutes it.

Limitations and Knowledge Gaps

1. **MLC2B-specific epidemiology is absent.** Prevalence, incidence, and sex distribution of the dominant subtype specifically are not quantified; most cohort data pool recessive MLC.
2. **Dominant-negative mechanism is partly inferred.** The cis/trans self-interaction model (F3) rests on structural modelling; direct structural (cryo-EM/crystallographic) confirmation of mutant GlialCAM self-association defects is still needed.
3. **No MLC2B-specific animal model fully captures remission.** Most genetic models are MLC1-null (recessive); dominant hepaCAM in-vivo work is recent and does not yet reproduce the human improving course.
4. **Molecular basis of remission is unexplained.** Why single-allele defects resolve developmentally (compensation? declining MLC1/GlialCAM expression? myelination maturation?) is hypothesized but not demonstrated.
5. **GPRC5B node is emerging.** Its weight relative to the ion-homeostasis defect, and whether it contributes to MLC2B specifically, is unresolved.
6. **No biomarkers or therapies.** No fluid biomarker, no disease-modifying treatment, and no natural-history registry data specific to MLC2B.
7. **Penetrance quantification is lacking.** "Incomplete penetrance/variable expressivity" is qualitative; no numeric penetrance estimates exist for specific dominant alleles.

Proposed Follow-up Experiments / Actions

1. **Natural-history / genotype–phenotype registry for HEPACAM-dominant MLC2B**, with serial head circumference, MRI (quantitative white-matter water content), and neurocognitive outcomes to quantify remission timing, penetrance, and expressivity per allele.
2. **Knock-in mouse (or iPSC-astrocyte/organoid) carrying a bona fide dominant MLC2B allele** (e.g., p.Q56P) to test whether it reproduces the **remitting** course and to dissect developmental compensation.
3. **Structural validation** (cryo-EM/AlphaFold-Multimer + biophysics) of GlialCAM Ig1 cis/trans self-association, comparing dominant vs recessive mutants, to confirm the dominant-negative surface model (F3).

4. **Interactome time-course** in developing astrocytes to test whether loss of Connexin 43/ClC-2 association and gain of KCNQ2 binding (F6) normalize as edema remits — linking molecular rewiring to clinical remission.
5. **VRAC/AQP4 functional assays** in MLC2B patient-derived astrocytes vs MLC2A, to determine whether the dominant single-allele defect produces a *partial* (recoverable) versus *complete* homeostatic failure, testing the allele-dosage model.
6. **Evaluate GPRC5B signaling** (gain-of-function) in HEPACAM-dominant backgrounds to define its contribution and druggability within the MLC pathway.
7. **Biomarker discovery** (CSF/plasma proteomics, imaging quantitation) to track and predict the remitting course and support any future trials.

Report compiled from 7 confirmed findings and 23 reviewed papers over 5 completed investigation iterations. Evidence types are annotated (human clinical, mouse model, in vitro, computational/structural). Causal gene throughout: HEPACAM/GlialCAM; disease: MLC2B (OMIM 613926, MONDO:0013491).
